

ORGANIC POLLUTANT MONITOR OPSA-150

Installation Manual

CODE:GZ0000054386

Preface

This manual describes the installation procedure of the ORGANIC POLLUTANT MONITOR, OPSA-150.

Be sure to read this manual before installation of the product. Also safely store the manual so it is readily available whenever necessary.

Product specifications and appearance, as well as the contents of this manual are subject to change without notice.

■ Warranty and Responsibility

HORIBA warrants that the Product shall be free from defects in material and workmanship and agrees to repair or replace free of charge, at HORIBA's option, any malfunctioned or damaged Product attributable to HORIBA's responsibility for a period of one (1) year from the delivery unless otherwise agreed with a written agreement. In any one of the following cases, none of the warranties set forth herein shall be extended;

- Any malfunction or damage attributable to improper operation
- Any malfunction attributable to repair or modification by any person not authorized by HORIBA
- Any malfunction or damage attributable to the use in an environment not specified in this manual
- Any malfunction or damage attributable to violation of the instructions in this manual or operations in the manner not specified in this manual
- Any malfunction or damage attributable to any cause or causes beyond the reasonable control of HORIBA such as natural disasters
- Any deterioration in appearance attributable to corrosion, rust, and so on
- Replacement of consumables

HORIBA SHALL NOT BE LIABLE FOR ANY DAMAGES RESULTING FROM ANY MALFUNCTIONS OF THE PRODUCT, ANY ERASURE OF DATA, OR ANY OTHER USES OF THE PRODUCT.

■ Trademarks

Generally, company names and brand names are either registered trademarks or trademarks of the respective companies.

Conformable Directive

This equipment conforms to the following directives and standards:



Directives: the EMC Directive 89/336/EEC, in accordance with Article 10 (1) of the Directive
the Low Voltage Directive 73/23/EEC
Standards: [the EMC Directive]
EN61326: A3: 2003 Class A, Industry
[the Low Voltage Directive] EN61010-1: 2001

● Installation Environment

This product is designed for the following environment.

- Installation Categories II
- Pollution degree 2

● Information on Disposal of Electrical and Electronic Equipment



The crossed out wheeled bin symbol shown on the product or accompanying documents indicates separate collection for waste electrical and electronic equipment (WEEE) under the WEEE Directive 2002/96/EC in the European Union.

This product should not be disposed of as unsorted household waste.



Your correct disposal of WEEE will contribute to reducing wasteful consumption of natural resources and protecting human health and the environment from potential negative effects caused by hazardous substances in products.

Contact your supplier for information on applicable disposal methods.

FCC Rules

Any changes or modifications not expressly approved by the party responsible for compliance shall void the user's authority to operate the equipment.

■ WARNING

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

For your safety

Warning messages are described in the following manner. Read the messages and follow the instructions carefully.

● Meaning of warning messages

 **DANGER**

This indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. This signal word is to be limited to the most extreme situations.

 **WARNING**

This indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

 **CAUTION**

This indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

Without safety alert indication of hazardous situation which, if not avoided, could result in property damage.

● Symbols



Description of what should be done, or what should be followed



Description of what should never be done, or what is prohibited

■ Safety Precautions

This section provides precautions to enable you to use the product safely and correctly and to prevent injury and damage. The terms of DANGER, WARNING, and CAUTION indicate the degree of imminency and hazardous situation. Read the precautions carefully as it contains important safety messages.

 WARNING	
	Do not open the cover inside the door of the control unit while the power is ON. Before opening the cover, unplug or turn off the power of the unit to avoid the danger of electric shock.
	Do not attempt to disassemble or modify the unit. Doing so could lead to bodily harm and/or injury or may cause damage to the unit or the sample.
	Maintain ground to avoid electric shock.

 CAUTION	
	The motor generates a high temperature. When replacing the components of the motor, disassemble the motor after it has cooled down sufficiently.
	Take care when handling the cell of the analyzer unit. The cell of the analyzer unit is made of glass. Handle with great care so as not to break the glass.
	Do not touch the cell or directly view the UV lamp of the analyzer unit when you check the unit operation.

Points of concern

Use of the equipment in a manner not specified by the manufacturer may impair the protection provided by the equipment. And it may also reduce equipment performance.

Exercise the following precautions:

- Use the unit in the temperature range shown in the general specifications.
- Use the unit in an environment free from corrosive gas.
- Avoid using the unit in a location close to equipment using strong electric power (such as an electric furnace), or near a radio or sound wave source.
- Do not give a shock or large vibrations to the unit. To move the unit to another location, contact our service or sales office.
- Avoid turning ON the power again immediately after turning it OFF.
- Do not press buttons using wet hands, or sharp objects such as pen tips or screwdrivers.

Disposal of the product

When disposing of the product, follow the related laws and/or regulations of your country for disposal of the product.

Description in this manual

Note

This interprets the necessary points for correct operation and notifies the important points for handling the unit.

Reference

This indicates the part of where to refer the information.

Tip

This indicates reference information.

Documents related to this product

The following documents are related to this product.

● Instruction Manual for OPSA-150

Contains information about necessary work and procedures for daily measurement operations.

● Installation Manual for OPSA-150 (this manual)

Contains information about the installation procedure.

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1 Overview

1.1 Cautions

This manual provides the procedure for assembling and installing the UV meter and pole stand, and cautions.

To obtain stable operation and reliable measurement of the UV meter for a long period of time, you need correct installation work from setup through piping.

If the meter is not installed correctly, unexpected problems may occur such as incorrect analyzer indications, frequent malfunctions, and increases in required inspection and maintenance.

Problems due to incorrect installation or use are not covered by the warranty. Be sure to fully follow the manual.

1.2 Composition

For OPSA-150 installation, both the UV meter itself and OPSA-150 pole stand are required. The following table shows the components included with the UV meter and pole stand.

Table 1 Components included with UV meter

Component	Name	Specifications	Quantity
Operation unit	OPSA-150 operation unit	Standard specification (outdoor case)	1
Analysis unit	OPSA-150 analysis unit		1
Overflow tank	Overflow tank	Float switch	1
	U bolt (large) set	M8, screw nut	2
	Pipe stay set	Pipe, U bolt, screw nut	1
	Hose set	1.2 m, with one-touch joint	1
Measuring tank and accessory	Measuring tank		1
	Blind plate		1
	Calibration tank		2
	V-4 pipe set	V-4 valve, pipe	1
	V-2 pipe set	V-2 valve, T-pipe	1
	Y-pipe set	Y-pipe, hose clip	1
	M8 bolt set	Used for attaching the measuring tank	4
	Accessories for analysis unit	Drying agent, calibrating solution, wiper, washer	1
	OPSA-150 Operation Manual	Standard specification	1

Table 2 Components included with the pole stand

Component	Name	Specifications	Quantity
Pole stand	OPSA-150 pole stand	Standard specification	1
Analysis unit mount	OPSA-150 analysis unit mount	Analysis unit mount	1
Operation unit bracket	Converter bracket	Converter bracket	1

2 Installation

2.1 Selecting an installation location

OPSA-150 is intended for indoor use. When installing OPSA-150, select a location that satisfies the requirements below.

2.2 Cautions for installation environment

2.2.1 Sample water condition

- Water temperature: 2°C to 40°C (must not be frozen)
- Water flow rate: 2 L/min to 20 L/min

2.2.2 Surrounding environment

- No direct sunlight to the meter
- Close to the water sampling point
- Close to a water pipe and drain facility/accessible to power supply
- No dust, mist or corrosive gas in the air
- Good ventilation
- Protected from rain and wind
- Surrounding temperature: 0°C to 40°C/relative humidity: 85% or less
- Stable location with few vibrations or shocks
- Altitude of 2000 m or less

2.3 Installing OPSA-150

2.3.1 Positions of foundation bolts for pole stand

Firmly attach M8 foundation bolts as shown below.

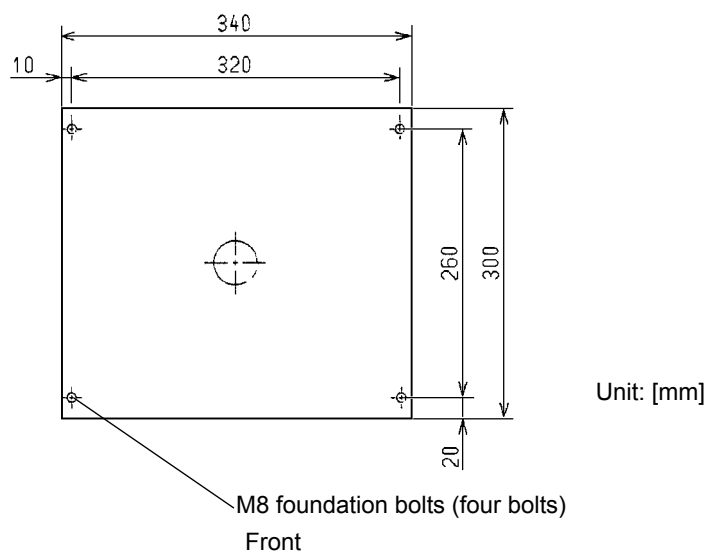


Fig. 1 Positions of the foundation bolts

2.3.2 Service and maintenance space

Prepare the space shown below for service and maintenance.

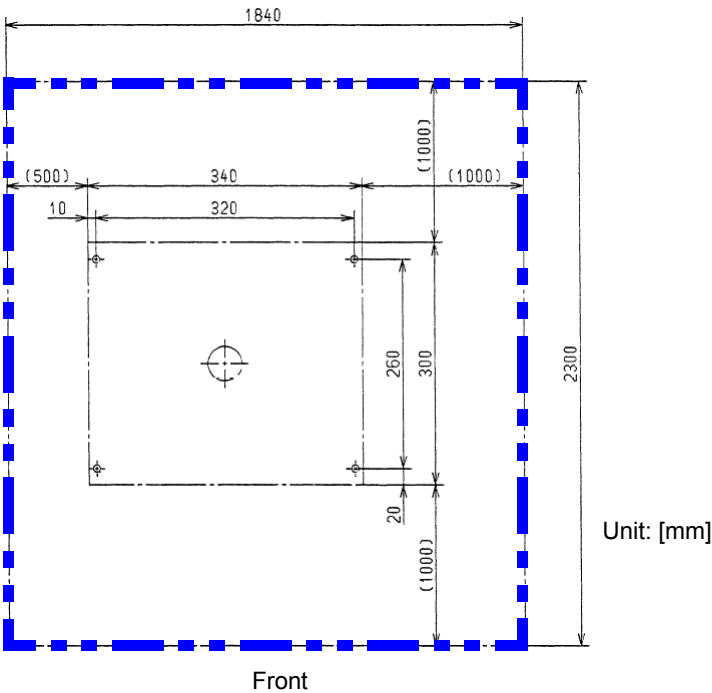


Fig. 2 Service and maintenance space

2.3.3 Figure of completed pole stand mount

The following figure shows the completed pole stand mount.

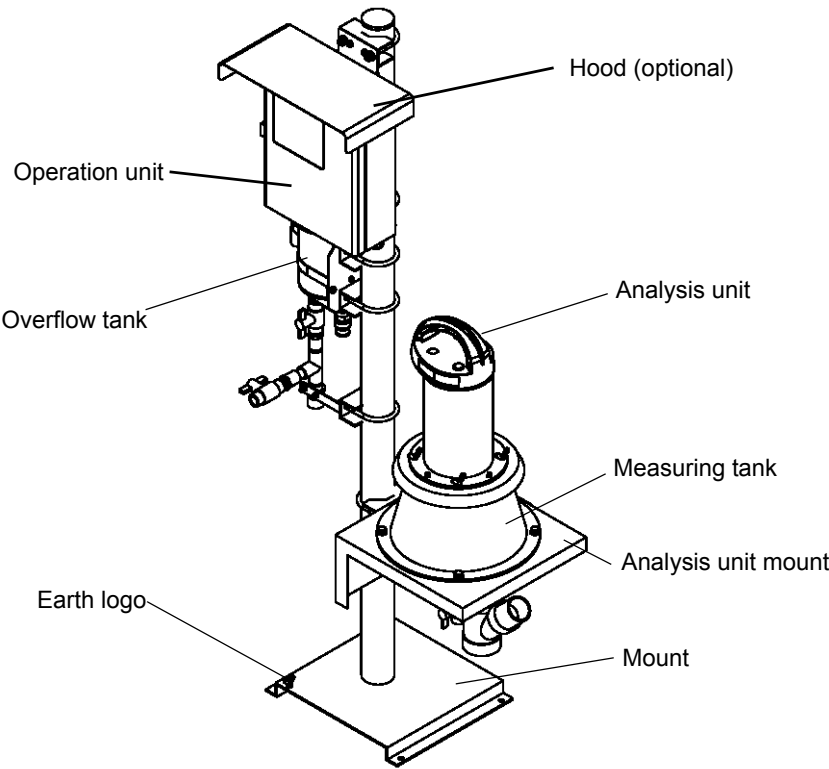


Fig. 3 OPESA-150 completed pole stand

2.3.4 Pole stand assembly

1. Use four M8 bolts to prevent the mount from moving.

Make sure to locate the earth logo on the mount at the front left (looking from the front).

2. Use two U bolts and attach the analysis unit mount to the pole as shown below.

Make sure that there is a 350 mm to 500 mm distance between the mount and floor. (450 mm is recommended.)

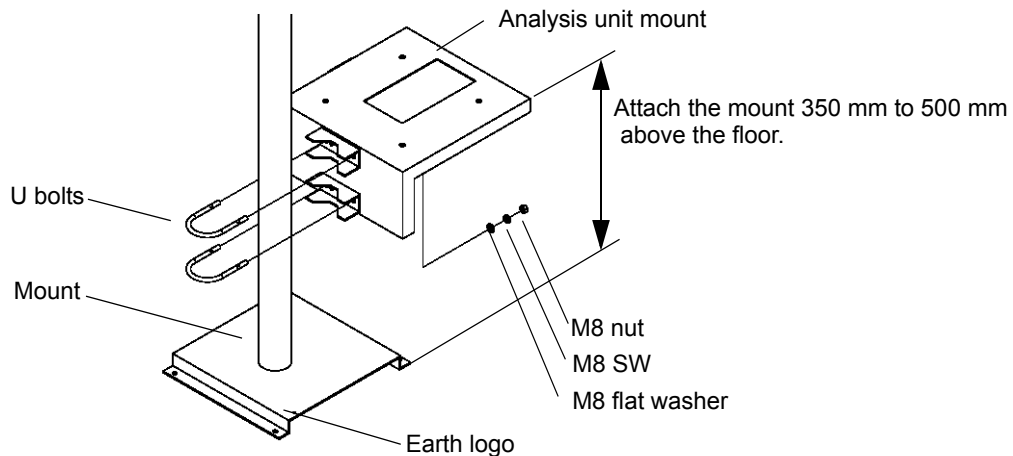


Fig. 4 Analysis unit mount installation

2.3.5 Measuring tank assembly

1. Attach the measuring tank included in the meter to the analysis unit mount using four M8 × 16 bolts.

Make sure to locate the thicker drain of the pipe on the right (looking from the front).

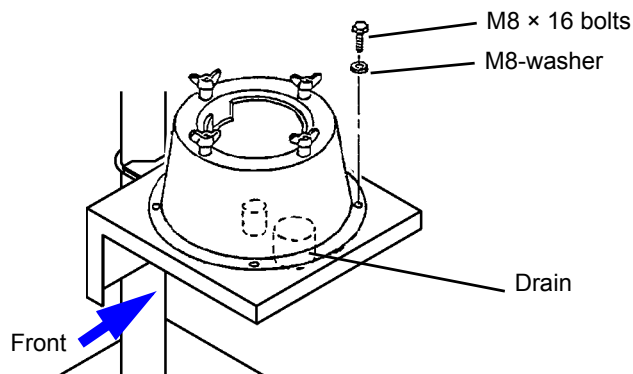


Fig. 5 Measuring tank assembly

2. Remove the four butterfly nuts, attach the blind plate and reattach the butterfly nuts.

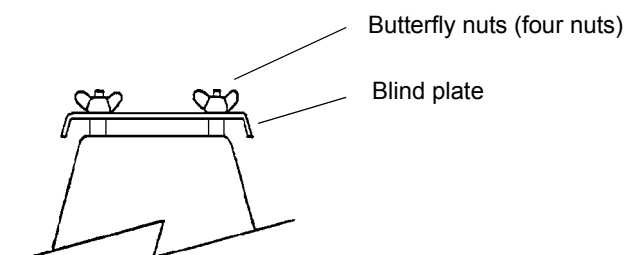


Fig. 6 Blind plate installation

3. Screw the pipe unit with V-4 valve attached into the bottom of the measuring tank.

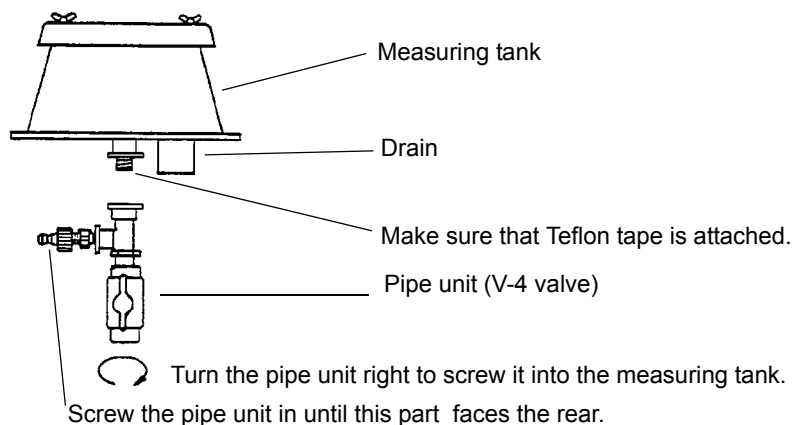


Fig. 7 V-4 valve installation

Note

Make sure that Teflon tape is attached to the screw on the bottom of the measuring tank before screwing in the pipe unit.

4. Insert the Y-pipe into the drain on the bottom of the measuring tank and attach it with the hose clip.

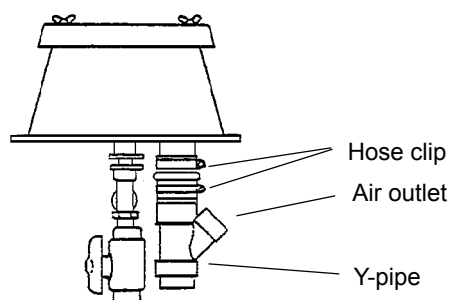


Fig. 8 Y-pipe installation

2.3.6 Overflow tank assembly

1. Screw the pipe unit with the valve V-2 attached into the valve V-1 valve on the bottom of the overflow tank.

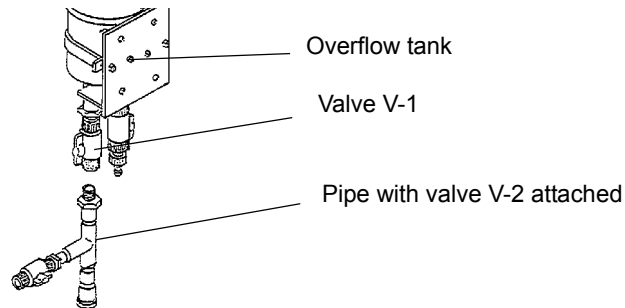


Fig. 9 Installation of valve V-2

2. Use U bolts (large/M8) to attach the overflow tank to the pole. Make sure that there is a distance of approximately 400 mm between the analysis unit mount and the bottom of the overflow tank.

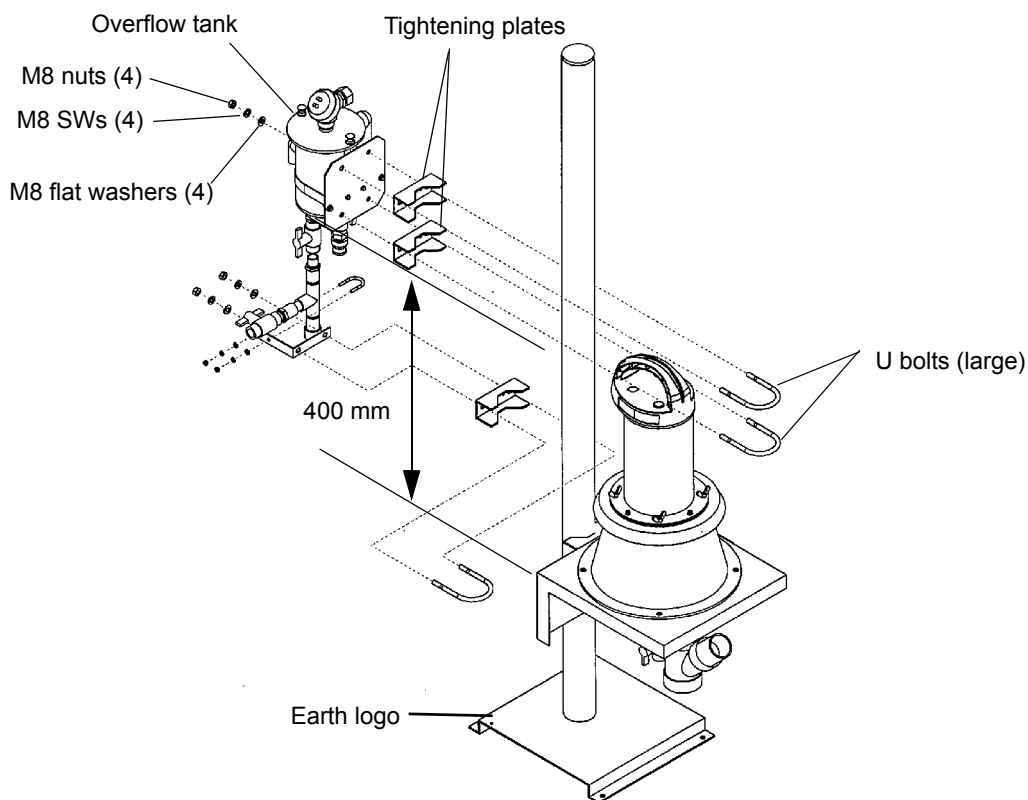


Fig. 10 Overflow tank installation

3. Attach the pipe stay to the pipe unit (V-2 valve attached) temporarily using a U bolt (small/M5), and then fasten the pipe stay to the pole stand using another U bolt (large/M8). After making sure that there is no excess pressure on the pipe unit, firmly fasten the U bolt (small/M5) attached first.

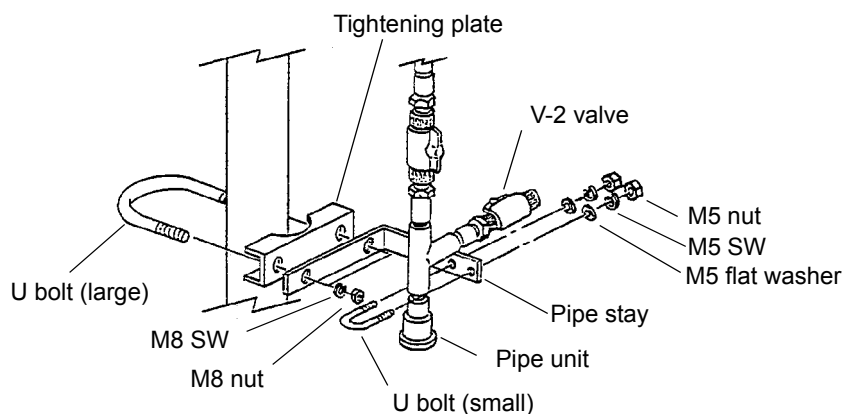


Fig. 11 Pipe stay installation

4. Connect the outlet hose end of the valve V-3 on the bottom of the overflow tank to the flexible tube that is attached to the hose end on the bottom of the measuring tank. One-touch joints are attached to both ends of the flexible tube. Push the joints together until they click.

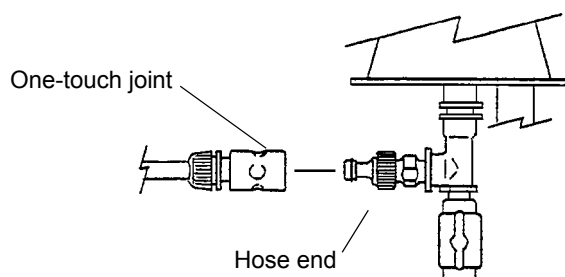


Fig. 12 One-touch joint installation

Note

To remove the flexible tube, pull the red part of the joint.

2.3.7 Operating unit assembly

1. Attach the converter bracket to the operation unit using four M6 hexagon head bolts. If you want to attach the optional hood, attach the components in this order: 1. operation unit; 2. converter bracket; 3. hood.
2. Attach the operation unit to the pole using U bolts (large/M8).

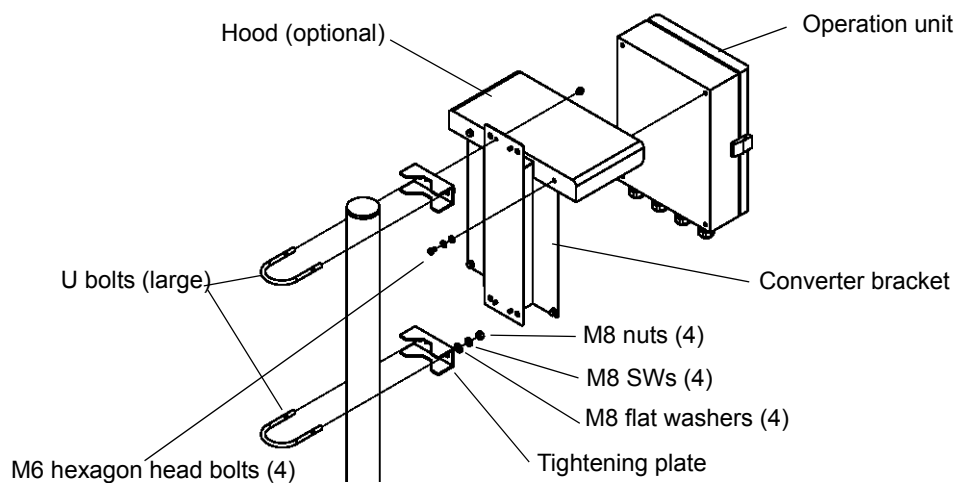


Fig. 13 Operation unit installation

3 Piping

3.1 Piping procedure

- When there may be pipe freezing, install a pipe warming device.
- To prevent backpressure, make the exit-side pipes as short as possible. In addition, the pipe exit must allow air to flow out freely, and must not be submerged in water. Do not install the pipe vertically.
- Before connecting the device, make sure to flush the sampling line.

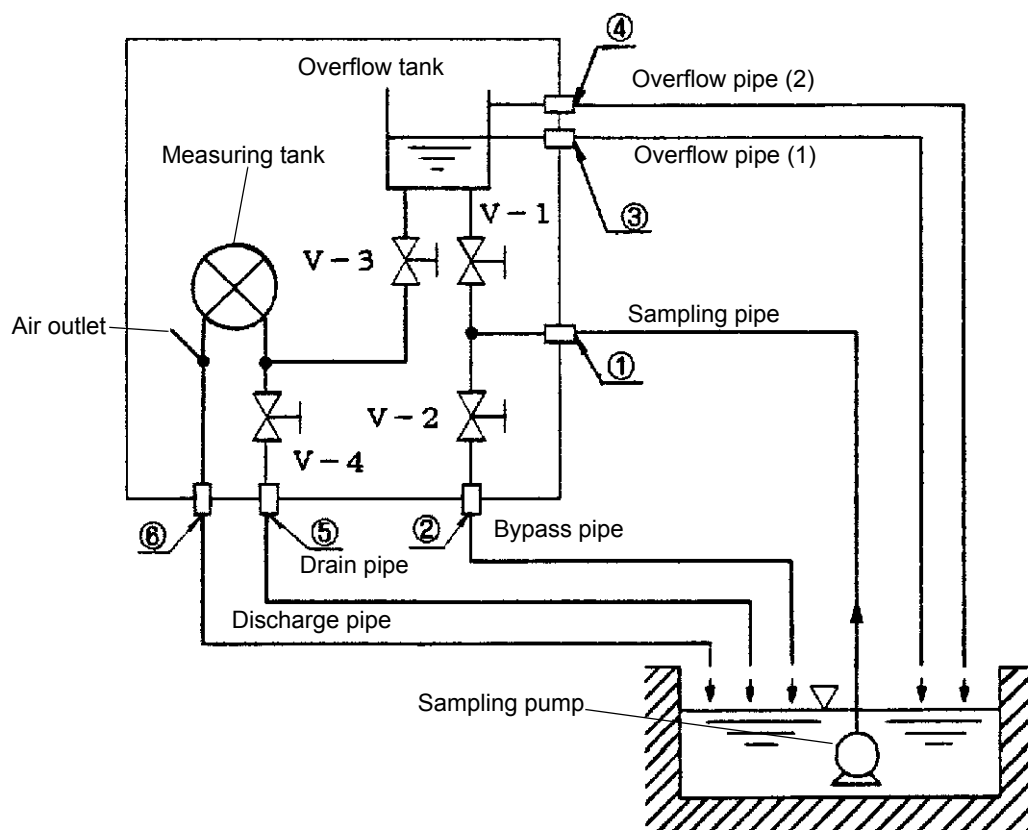


Fig. 14 Piping

- Refer to the following table for diameters of the pipes and the pipe connectors.

Table 3 Diameters of pipes and pipe connectors

Connector name	Pipe connector	Nominal pipe diameter (When a standard hard vinyl chloride pipe is used)
Sample water inlet	Rp-1/2 socket	13 mm or larger
Bypass outlet	Rc-1/2 socket	13 mm or larger
Overflow outlet (1)	Port diameter 13A socket	13 mm or larger
Overflow outlet (2)	Port diameter 20A socket	20 mm or larger
Drain outlet	Rc-1/2 socket	13 mm or larger
Discharge outlet	Port diameter 50A socket	50 mm or larger

Note

- The diameters of the pipes shown in Table 3 match the diameters of connectors for OPSA-150. If the water sampling point is distant from the device, use pipes with larger diameters than those mentioned in Table 3.
 - When piping the drain pipe, overflow pipes (1) and (2), and the discharge pipe, slope them sufficiently so that the ends of the pipes can not be submerged in water.
 - When using a flexible vinyl tube pipe, use a pressure-proof vinyl tube containing blade.
 - Removable piping is recommended so that the pipes can be disconnected for periodic cleaning.
-

4 Wiring

4.1 Wiring procedure

4.1.1 Earth connection

- Connect the earth according to Type D earth connection procedures.
- Be sure to connect the earth bolt.

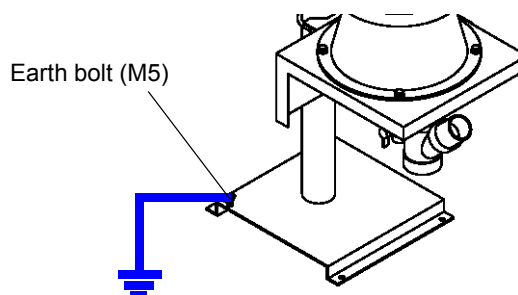


Fig. 15 Earth connection

Note

To prevent electric shocks, ensure correct earthing.

4.1.2 Power supply

Connect the power supply of the operation unit according to the signal table.

- Use a power cable of 1.25 mm² (AWG 16) or more in size.
- The surge absorbers in the power terminal block are connected in parallel to prevent noise. Tighten them together with the power cable.
- The earth must be connected for safe operation.
- Be sure to connect the power terminals in the order of L phase, E (earth) phase and N phase from the left. Ensure correct connection.

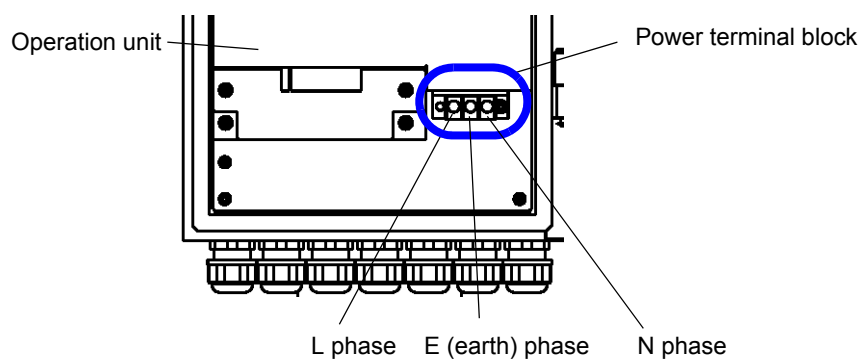


Fig. 16 Power terminal connection

4.1.3 Cable connection for analysis unit

- Pass the connector of the analysis unit through the cable clamp from the bottom left of the operation unit.
- Connect the connector of the analysis unit to the connectors CN 1 and CN 2 located in the bottom left of the inside of operation unit.
- Attach the round shield terminal to the screw hole under the connectors using a screw.

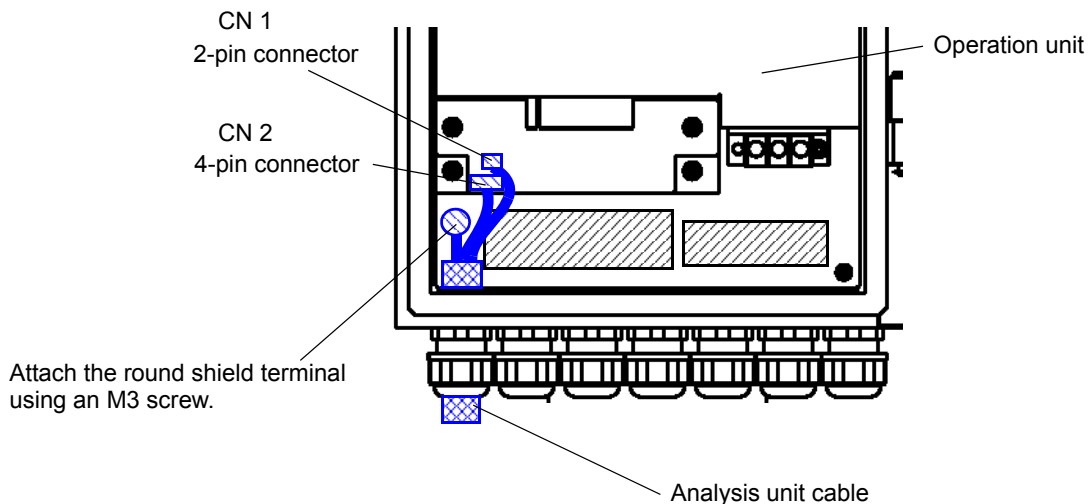


Fig. 17 Cable connection for analysis unit

4.1.4 Float switch cable connection for overflow tank

- Pass the float switch cable of the overflow tank through the cable clamp from the bottom of the operation unit.
- Connect cables to the No. 7 and 8 terminals on the power terminal block in the center bottom of the operation unit.

4.1.5 Signal line connection

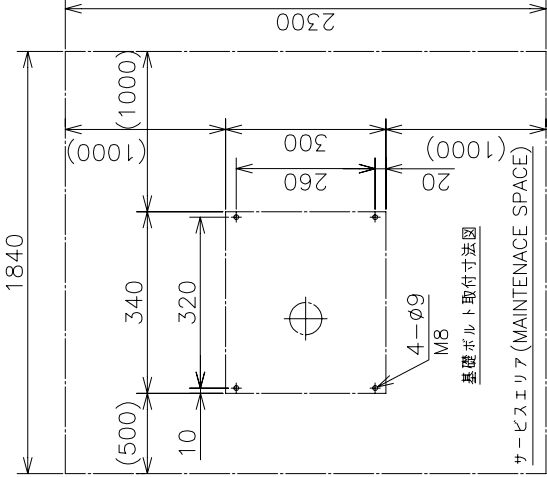
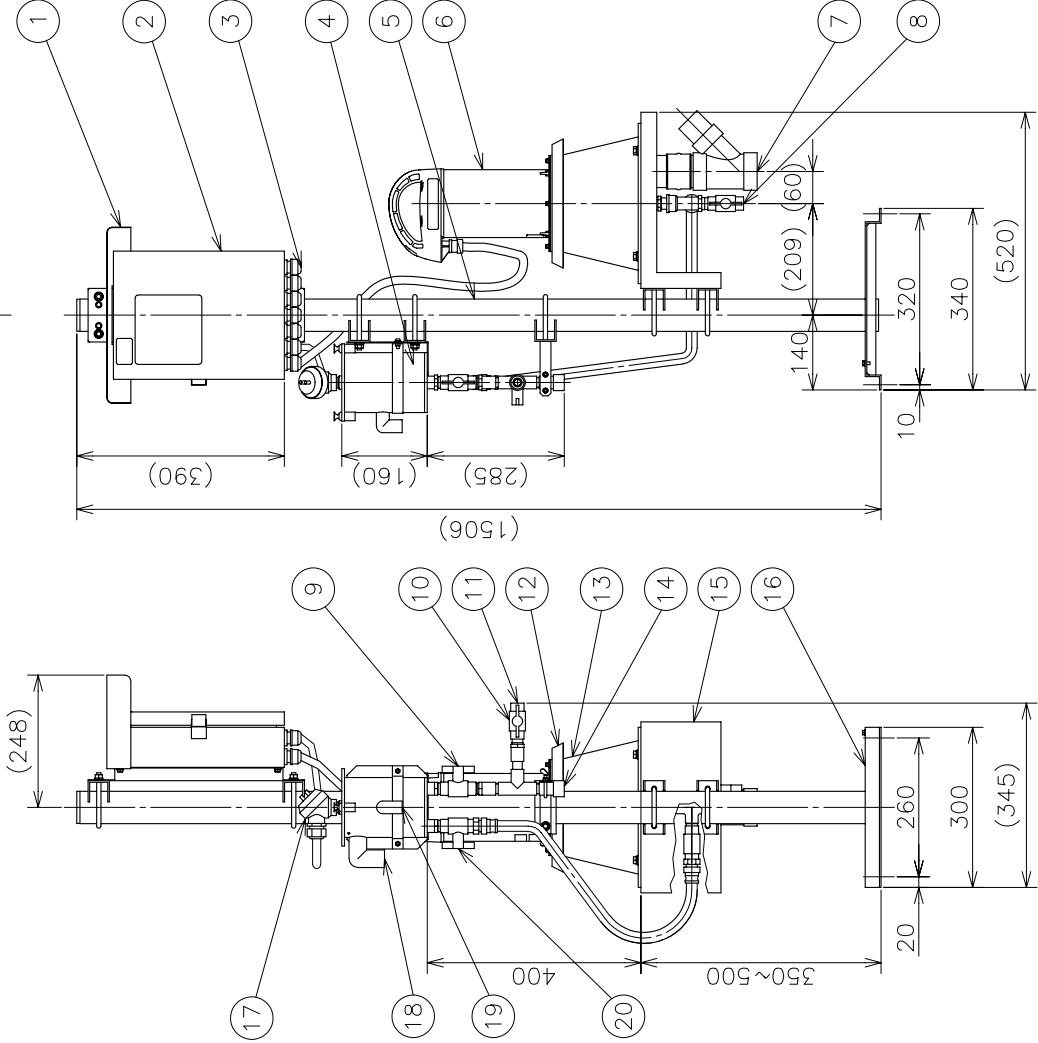
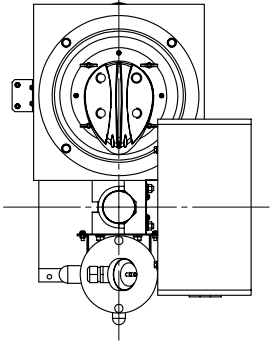
- Connect the signal lines according to the external connection-wiring figure (V1039557A).
- Use a 2-pin shield cable for the current power output signal line and earth the shield at the receiver side.
- Connect such devices as a surge absorber and noise killer in parallel to prevent noise in contact output signals.

Note

- Refer to the separate OPSA-150 Operation Manual.
- Do not connect the power cable to the signal line wiring section. Doing so will cause malfunctions.
- Do not turn the power on before introducing water to the system. The cell will rotate without water, which can cause malfunctions.

5 Related Figures

Name	Specifications	Figure No.
Dimensional outlines	Pole stand set	V3003480A
	Operation unit	V3001112
	Analysis unit	V1040265A
Flow schematic		V3001430B
External wiring diagram		V1039557A



正面 (FRONT)

注記
1.寸法公差は JIS B 0405 公差等級Cに準じます。

NOTE
1.TOLERANCE JIS B 0405 C

NO.	PARTS NAME	NOTES
1	屋外フード	
2	操作部本体	
3	配線取合口	
4	ワイリングポート	
5	オーバーフロー罐	
6	オーバーフロータンク	
7	ボール	
8	POLE	
9	分析部本体	
10	ANALYZER UNIT	
11	排水口	
12	EXHAUST PORT	50A SOCKET
13	ドレン出口 (V-4)	
14	DRAIN OUTLET	Rc1/2
15	ボールバルブ (V-1)	
16	BALL VALVE (V-1)	
17	ボールバルブ (V-2)	
18	BALL VALVE (V-2)	
19	バイパス出口	
20	BYPASS OUTLET	Rc1/2
21	ブラインド板	
22	BLIND BOARD	
23	測定槽	
24	MEASUREMENT TANK	
25	試料入口	
26	SAMPLE INLET	Rp1/2
27	分析台	
28	MEASUREMENT BASE	
29	ベース	
30	BASE	
31	フロートスイッチ	
32	FLOAT SWITCH	

NO.	SPEC.	PART NO.	REMARKS
NAME OPSA-150 屋外ボールスタンド 外形寸法図			
Revision of comments H.Yagi APR.20,2006			
REVISION DATE APR. 20, 2006 DESIGNED H.Yagi CHECKED K.Terada APPROVED O.Takiguchi DATE APR. 20, 2006 SCALE 1:10 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MILLIMETERS/THIRD ANGLE PROJECTION			
GZ0000032046A			
V3003480A			

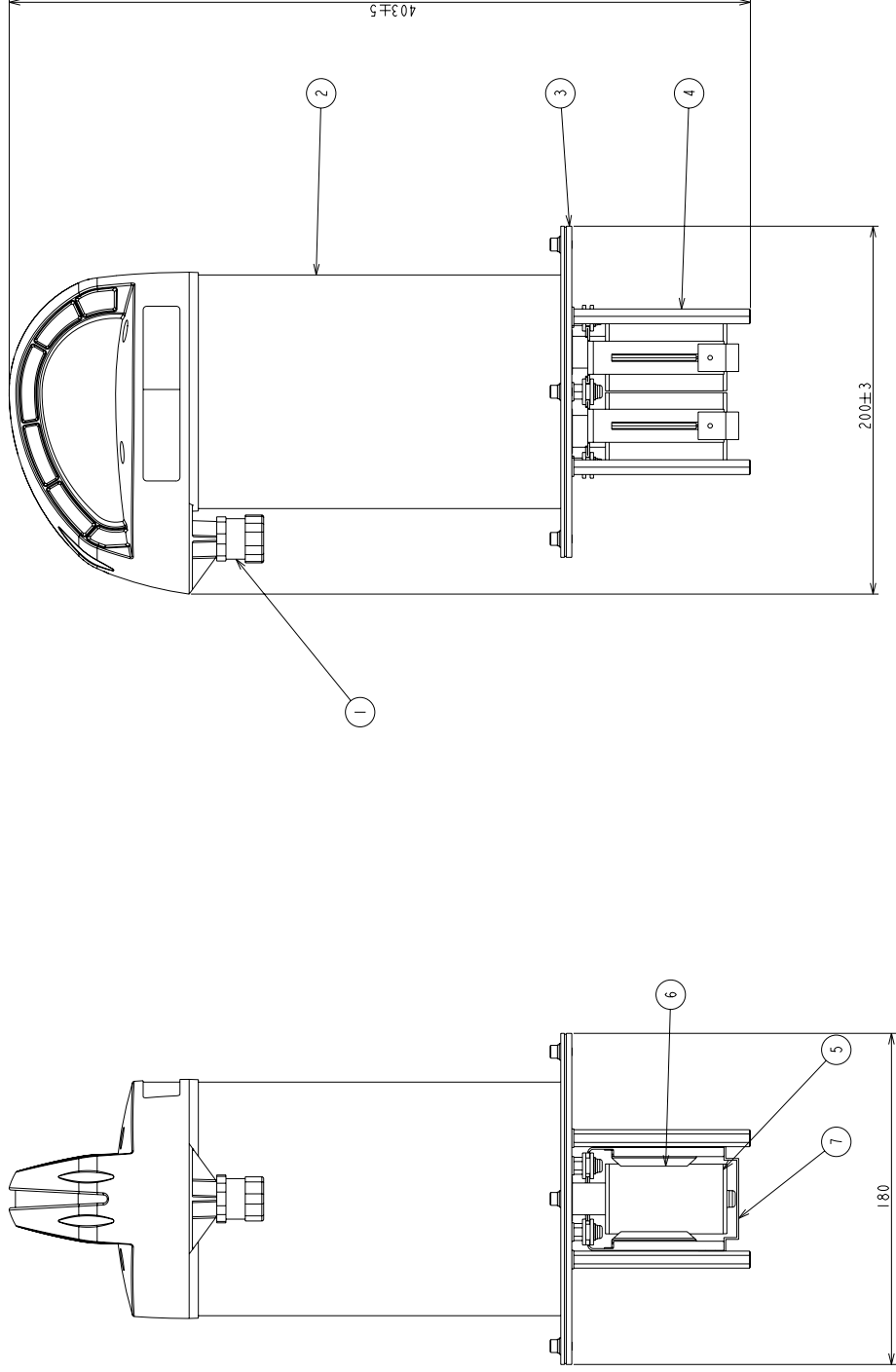
NO.		PARTS NAME	NOTES
1	A	取付板	SUS304
		FITTING PLATE	
2		操作部	
		OPERATING SECTION	
3	B	パチン錠	SUS304
		SNAP FASTENER	
4		配線口	O.D.φ6~φ11CABLE
		CABLE GRAND	
5	C	扉	SUS304
		DOOR	
6		ケース	SUS304
		CASE	
7	D	Uボルト	SUS304 M8
		U-BOLT	
8		締付座	SUS304
		CLAMP	
9	E	カバー(オプション)	SUS304
		COVER(OPTION)	

注記
1.塗装色 マンセル 5PB 8/1 全ツヤ
2.質量 約5kg
3.寸法公差は JIS B 0405 公差等級Cに準じます。

NOTE
1.SURFACE MUNSELL 5PB 8/1 GLOSS
2.WEIGHT APPROX. 5kg
3.TOLERANCE JIS B 0405 C

				NO.		SPEC.		PART NO.		REMARKS	
				NAME OPSA-150 外形寸法図 DIMENSIONAL OUTLINE							
				SCALE 1:5		UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MILLIMETERS/THIRD ANGLE PROJECTION					
DATE		JAN. 23, 2006		DATE		JAN. 24, 2006					
DRAWN		DESIGNED H.Yagi		CHECKED N.Yoshioka		APPROVED		K.Ogura			
								GZ0000027811			
								V3001112			

No.	PARTS NAME	REMARKS
1	Cable conduit	
2	Case	
3	Base	
4	Base leg	
5	Cell	
6	Wiper	
7	Wiper fixture	

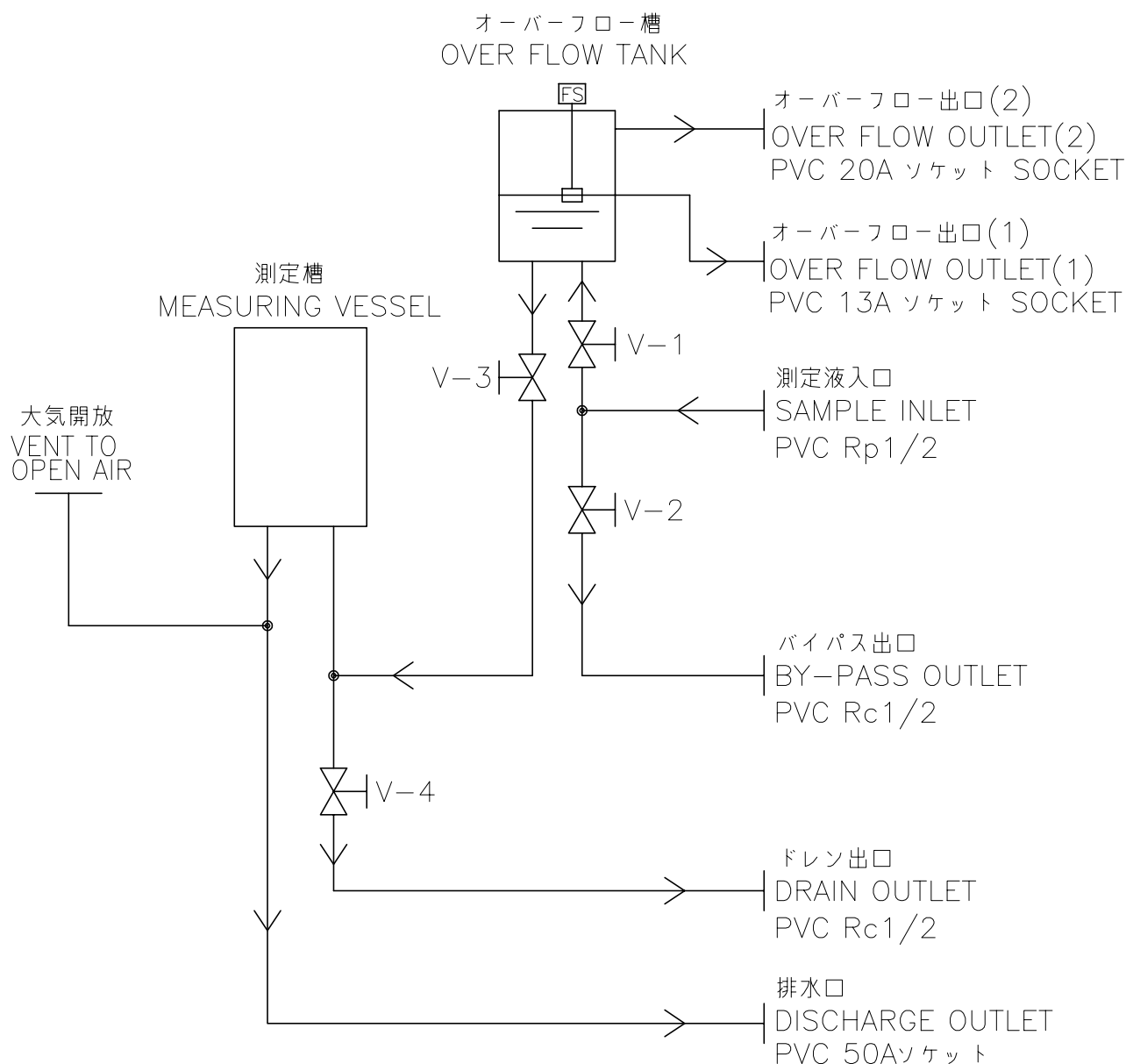
[illegible]

V3001430B

GZ0000015945B

HORIBA GROUP

EXTERIOR DRAWING



注記：排水口は背圧がかからない様に配管してください。

COMMENT

Sample outlet must be free from pressure which disturbs smooth drainage.

The drained water should be free fall at the end of the drain pipe.

				NO.	SPEC.	PART NO.	REMARKS
				NAME OPSA-150 フローシート FLOW SCHEMATIC			
▽	Revision of comments	H.Yagi	APR.20,2006				
▽	Addition of English description	H.Yagi	NOV.17,2005				
REVISED				SCALE	FREE	UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MILLIMETERS/THIRD ANGLE PROJECTION	
DATE	DATE APR. 20, 2006	DATE APR. 20, 2006	DATE APR. 20, 2006	GZ0000015945B			
DRAWN	DESIGNED H.Yagi	CHECKED K.Terada	APPROVED O.Takiguchi	V3001430B			

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